





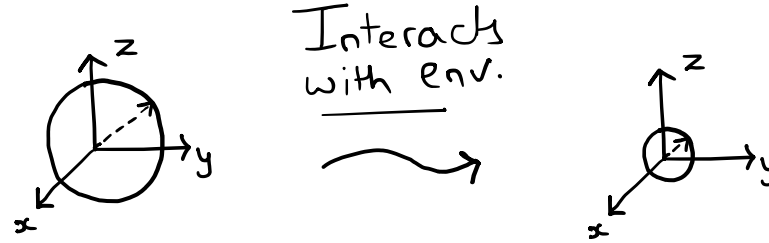
Quantum Error Mitigation(I)

Bit Flip Error Mitigation

Pinakin Padalia

What are we covering in this module?

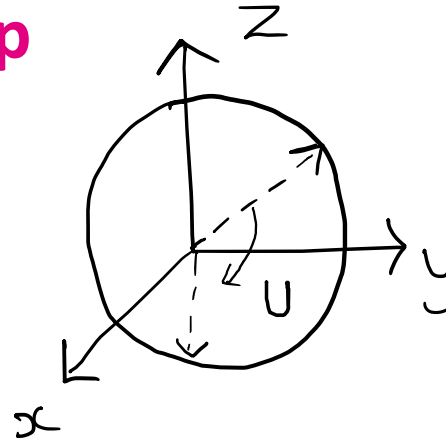
- Noise models
- **Quantum error mitigation**
- Quantum volume and performance matrices
- Hybrid quantum classical computing



What are we covering in this chapter?

- **Bit Flip Error Mitigation:**
 - Errors in Quantum: Recap
 - Strategy for error mitigation
 - Error Mitigation steps: 3 – qubit bit flip code
 - Limitations
- Phase Flip Error Mitigation
- Shor code (9 qubit code) and fault tolerance

Errors in Quantum: Recap

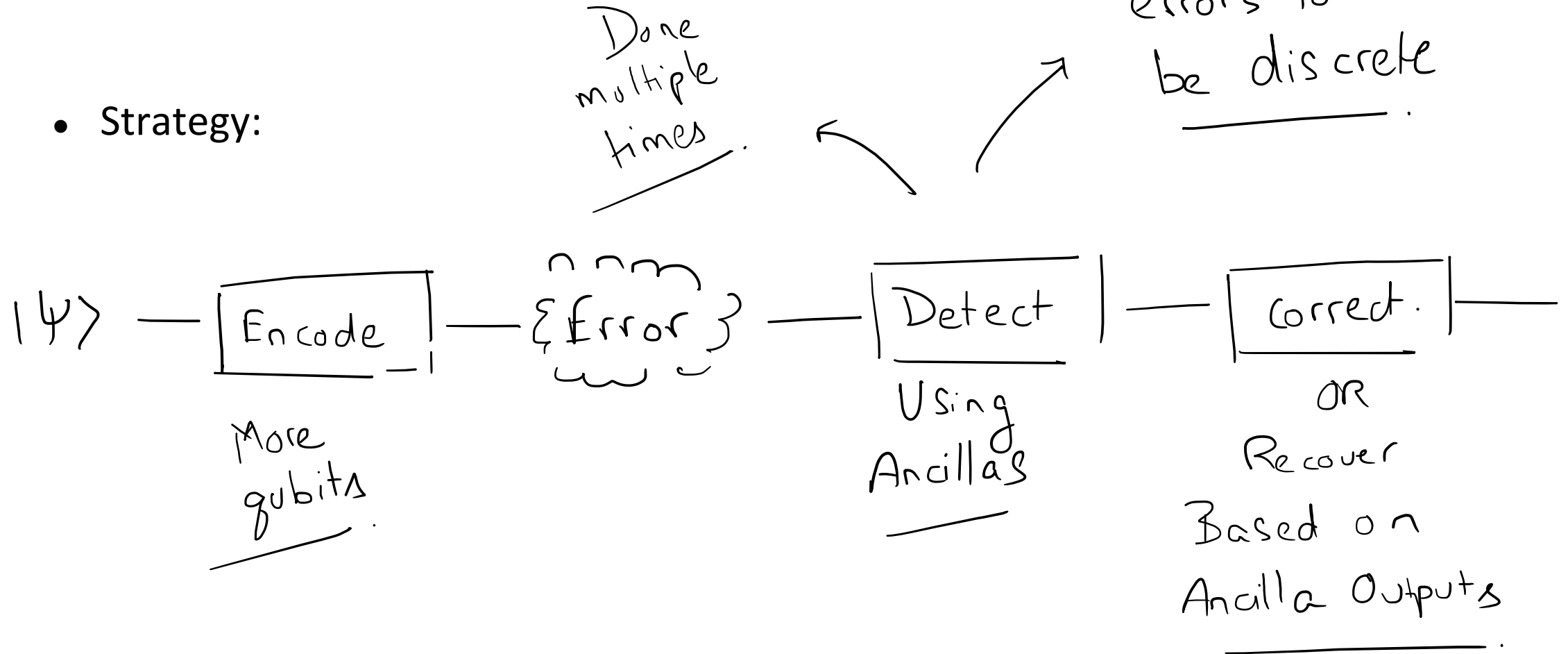


- Unitary operations & errors:
- Non-unitary operations & errors: 
- In general: errors take you from pure state to a mixed state

- Hurdles:
 - No cloning theorem — no copy allowed for ψ
 - Errors are continuous in nature — unlike classical.
 - Measurement destroys the state — so, no measure & restore

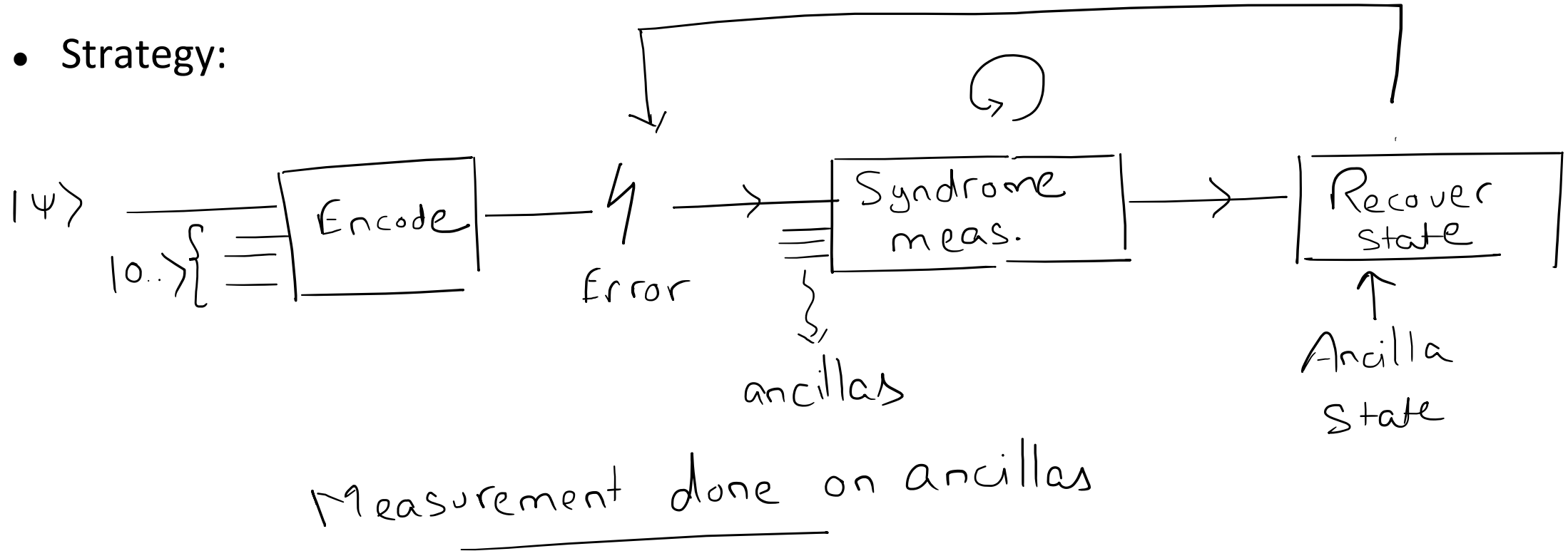
Strategy for error mitigation

- Strategy:



Strategy for error mitigation

- Strategy:



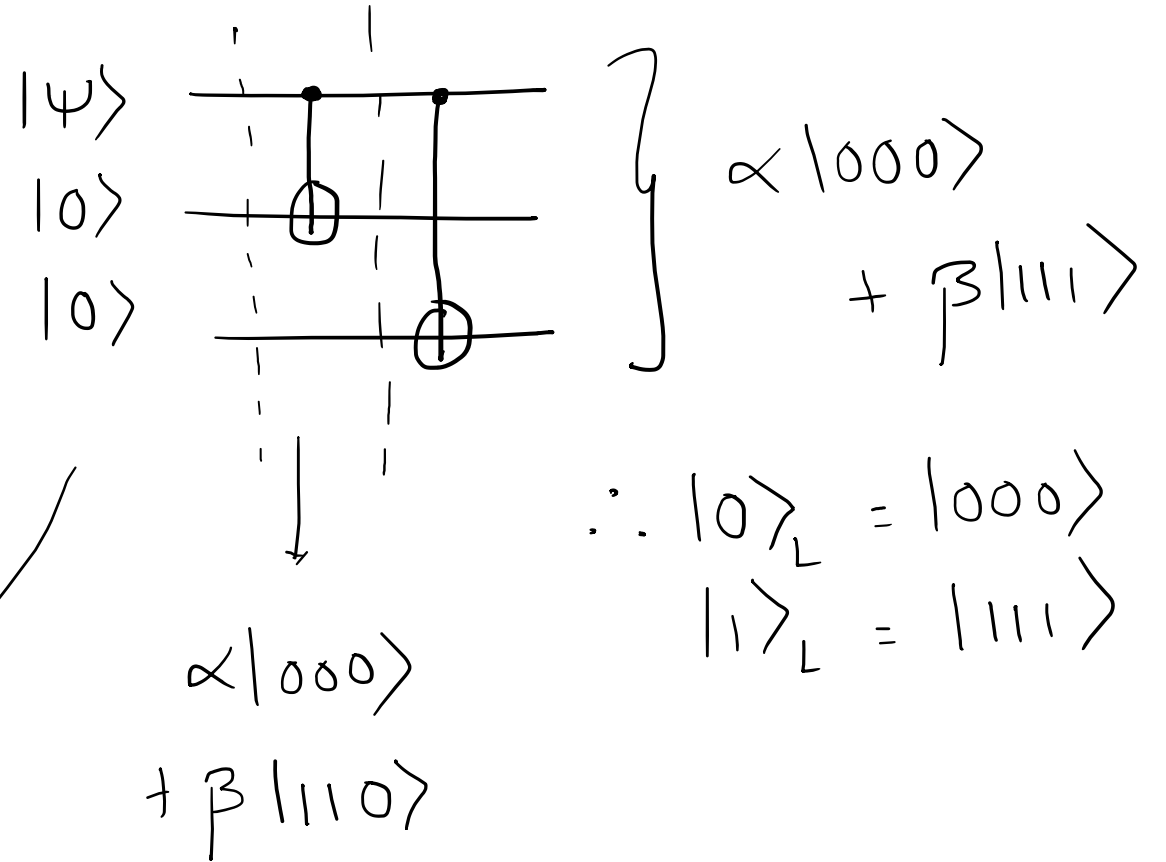
Bit Flip Error Mitigation

3-Qubit Bit Flip Code

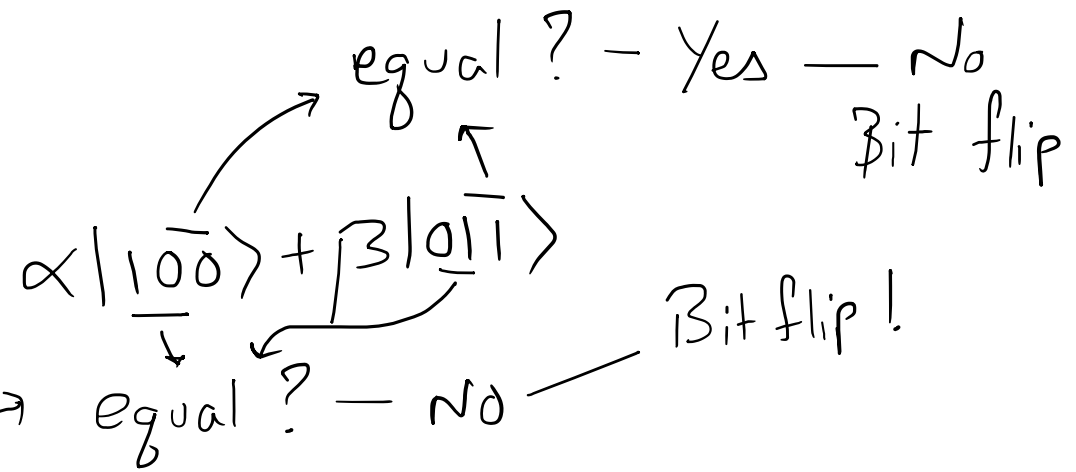
- Bit flip error: Pauli X error
- Mitigation: 3 – qubit bit flip code
 - Step 1: Encode

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\alpha|000\rangle + \beta|100\rangle$$



- Step 2: Add ancillas

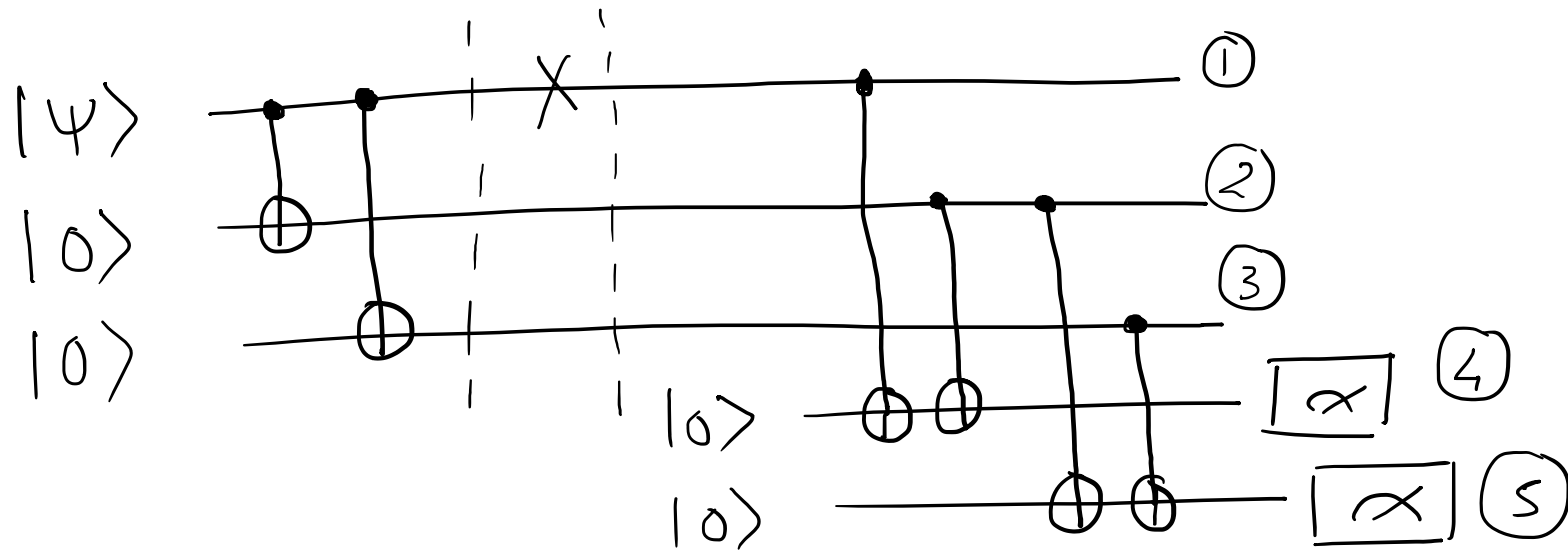


$$\alpha |100 \underline{00}\rangle + \beta |011 \underline{00}\rangle$$

Bit Flip Error Mitigation

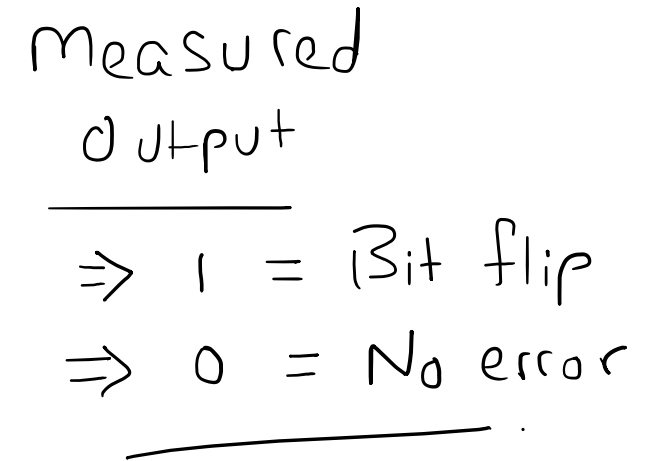
- Bit flip error: Pauli X error
- Mitigation:
 - Step 3: Syndrome Measurement

$\alpha|1\bar{0}\bar{0}\rangle + \beta|0\bar{1}\bar{1}\rangle$
 equal? — Yes — No
 Bit flip
 equal? — No — Bit flip!



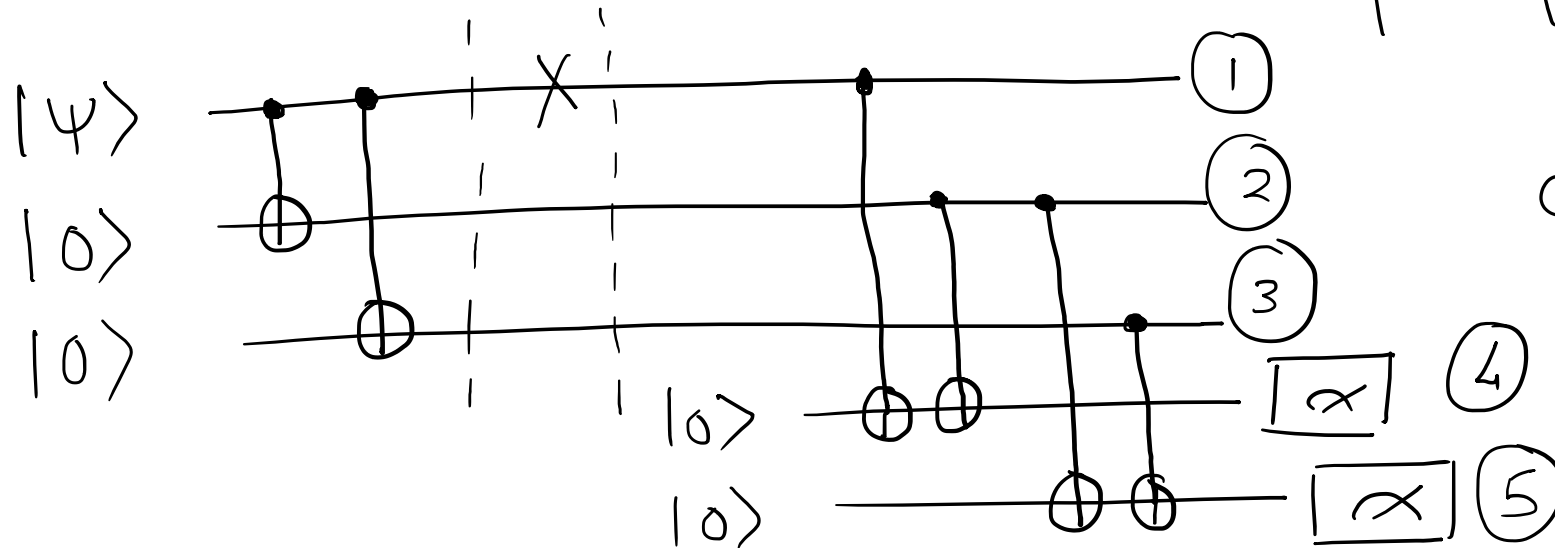
①	②	④
0	0	0
0	1	1
1	0	1
1	1	0

- $\alpha | \underline{1} \underline{0} \overline{0} \rangle + \beta | \underline{3} | \underline{0} \underline{1} \overline{1} \rangle$
 equal? — Yes — No
 Bit flip
 equal? — No — Bit flip!



Bit Flip Error Mitigation

- Bit flip error: Pauli X error
- Mitigation:
 - Step 4: Recovery



(4)	(5)
0	0
1	0
0	1
1	1

→ No error

→ error in (1)

→ error in (3)

→ error in (2)

↓
apply X gate to
relevant qubit

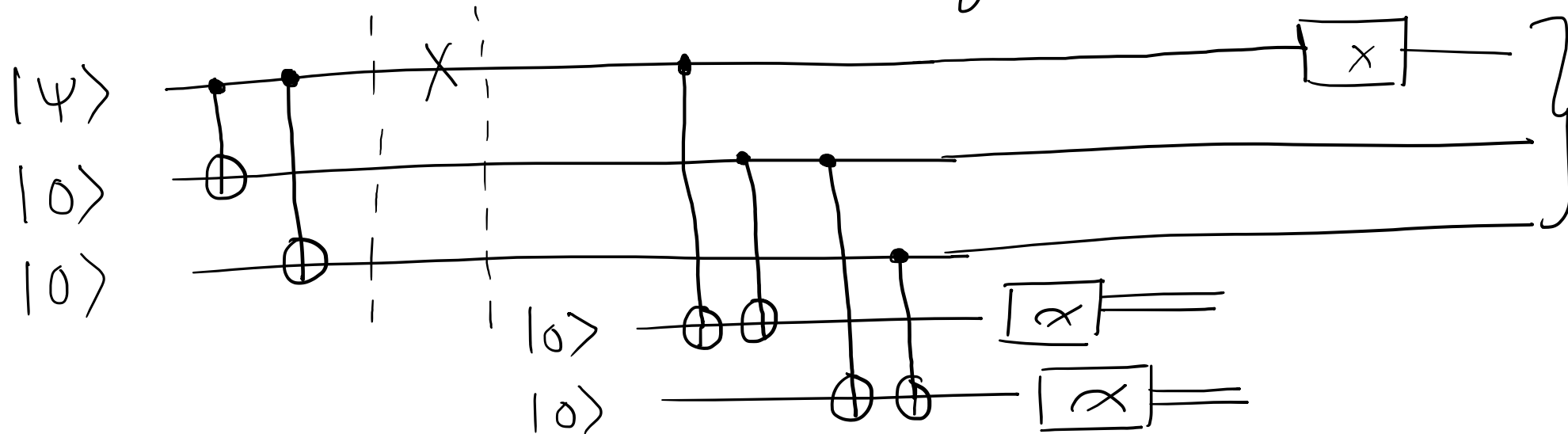
Bit Flip Error Mitigation

- Bit flip error: Pauli X error
- Mitigation:
 - Step 4: Recovery

④ ⑤
1 0

↓
Apply X gate
to qubit ①

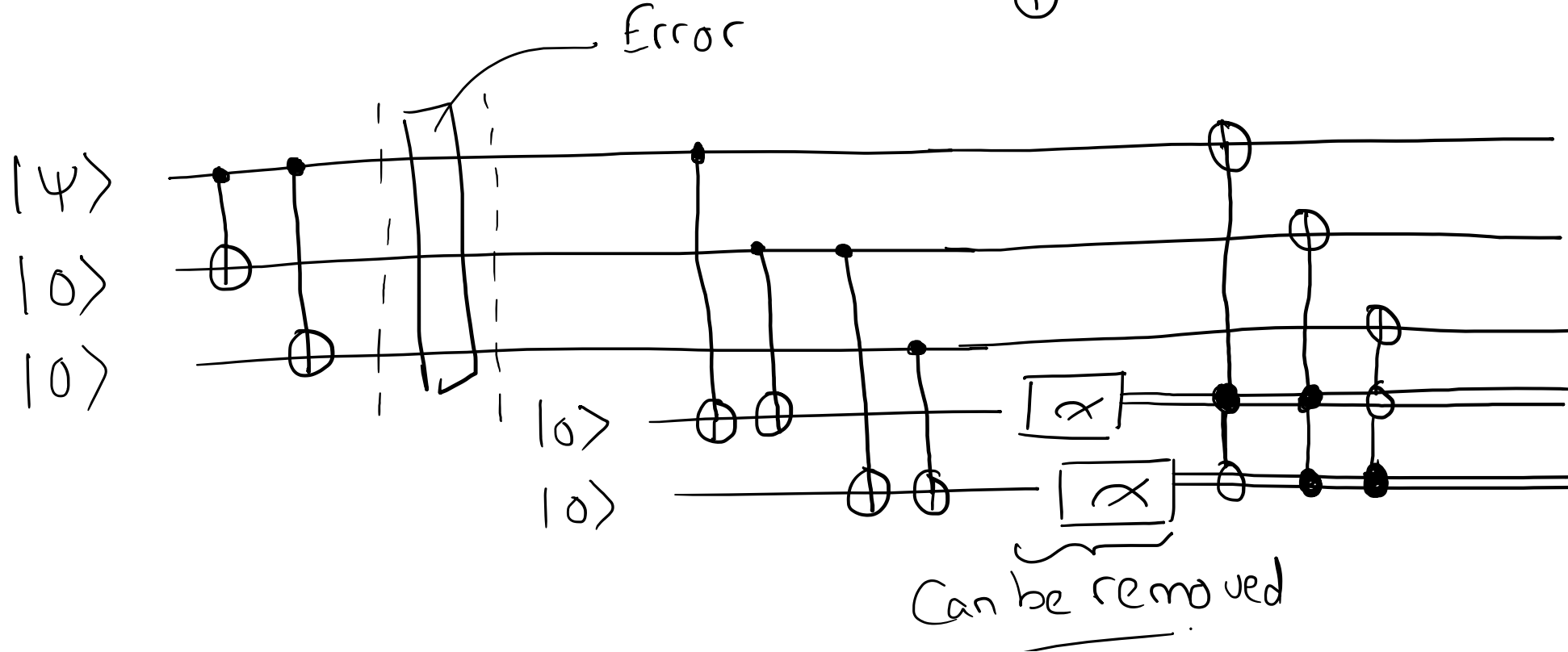
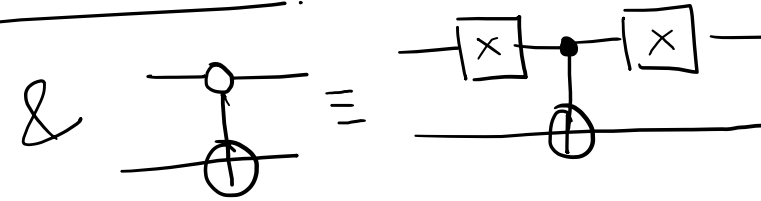
$\propto |000\rangle$
 $+ \beta |111\rangle$



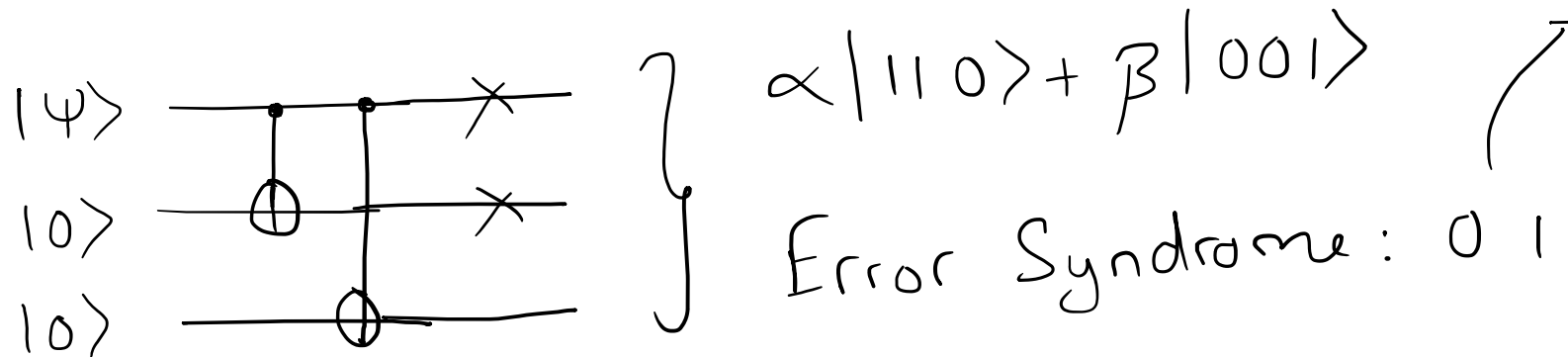
Bit Flip Error Mitigation

- Bit flip error: Pauli X error
- Mitigation: generalized picture

Controlled X $\equiv$ Controlled No T



- Bit flip error: Pauli X error
- Mitigation: Limitations
 - Requires more qubits – a scarce resource!
 - Requires repeated operations – problem with short coherence times!
 - Not effective for more than one error



Correct qubit 3!



Thank you for your attention!

- Michael A. Nielsen and Isaac L. Chuang. 2011. Quantum Computation and Quantum Information: 10th Anniversary Edition (10th. ed.). Cambridge University Press, USA.
- M. Wilde, "From Classical to Quantum Shannon Theory", Arxiv.org, 2019. [Online]. Available: <https://arxiv.org/pdf/1106.1445.pdf>.
- J. Preskill, "Lecture Notes for Physics 229: Quantum Information and Computation", Lorentz.leidenuniv.nl, 1998. [Online]. Available: https://www.lorentz.leidenuniv.nl/quantumcomputers/literature/preskill_1_to_6.pdf.